



NAUVAL ZULFIKAR

Applied ML / NLP Engineer · Responsible AI candidate Bandung, Indonesia (relocatable to Vienna, between June and October 2026) zulfikar.nauval1998@gmail.com · +44 7300 469048 · +62 821 2567 2264 [LinkedIn](#) · [GitHub](#) · [Portfolio](#) **Visa status:** UK Graduate Visa (active). Indonesian passport. Eligible for relevant research/student visas.

PROFILE

MSc Business Analytics graduate from Aston University with First Class honours, specialising in machine learning and NLP applications for responsible AI. Dissertation supervised by Dr. Ammar Al-Bazi focused on blockchain-integrated LLM analysis for supply-chain customer reviews. Proven track record in data-centric AI, AI safety, and efficiency, aligning with Prof. Dr. Martin Pawelczyk's research themes at Universität Wien.

EDUCATION

Aston University, Birmingham, UK (*MSc Business Analytics, 1:1 First Class · EQF Level 7 · Sep 2023 – Sep 2024*)

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**)

Ritsumeikan Asia Pacific University, Beppu, Japan (*BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019*)

- APU 50% Tuition Reduction Scholarship + JASSO Scholarship
- Distinctions: Business Data Analysis, Consumer Behaviour, Marketing Research

Indonesia AI (*NLP Engineer Training · EQF Level 5 · May – Nov 2023*)

- Contributed to the *2024 Indonesia Election Aspect-Based Sentiment Analyser* project
- Distinctions: Python Programming, Text Summarisation, Voice Recognition

Qarir Generator (Dubai, UAE) (*Data Science Global Accelerator · EQF Level 5 · Apr – Jun 2023*)

- Led sentiment-analysis development for the *Pfizer Public Sentiment Analysis* project
- Distinctions: Python, SQL, Pandas/NumPy

RESEARCH-RELEVANT PROJECTS

LLM-Generated Adaptive Shipper Decision Rules (2026)

- Developed an end-to-end RAG system using gpt-4o-mini for decision rules based on behavioural literature.
- github.com/nauvalZulfikar/LLM-Shipper-Profiles

Mini Digital Twin for Integrated Passenger-Freight Transport (2026)

- Built a three-layer Digital Twin prototype with live dashboards and interactive disruption-injection lab.
- github.com/nauvalZulfikar/Mini-Digital-Twin

VRPOD MILP Optimisation (2026)

- Implemented MILP optimisation for VRPOD, achieving 38.5% cost reduction on a toy instance.
- github.com/nauvalZulfikar/VRP-MILP

Crowd-Shipping ABM Behavioural Sweep (2026)

- Conducted a 450-run parameter sweep, demonstrating significant supply effect on acceptance rates.
- github.com/nauvalZulfikar/Crowd-Shipping-ABM-

INDUSTRY EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung (*Data Scientist & Project Manager · Sep 2022 – Present*)

- Automated ETL pipelines and dashboards, achieving 45% faster project delivery and 20% data-efficiency gain.

Syncwell, Birmingham, UK (*Marketing Data Scientist · Jul – Sep 2024*)

- Implemented predictive ML strategies, resulting in a 40% CTR increase and 22% follower growth.

PCOS Challenge, Birmingham, UK (*Strategic Data Scientist Specialist · Jun – Aug 2024*)

- Developed customer attribution models, reducing manual workload by 80%.

Bank Muamalat Indonesia, Jakarta (*Lead Data Scientist · Nov 2019 – Aug 2022*)

- Developed NLP models for credit analysis, achieving a 5.3% investment return.

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Working paper on ML regressor evaluation for retail sales forecasting.

TECHNICAL SKILLS

ML/NLP: PyTorch, HuggingFace Transformers, DeBERTa-v3 fine-tuning, gpt-4o-mini API, RAG, sentence-transformers

Languages/Infrastructure: Python, SQL, R, PySpark, FastAPI, Streamlit, AWS, Docker, Git

Mathematical/Formal: MILP, Agent-Based Modelling, Discrete-Event Simulation, Bayesian methods

Governance: Model validation, reliability, audit-grade documentation, A/B testing

LANGUAGES

Indonesian (*Native*) · **English** (*Fluent, IELTS 7.0 overall*) · **Japanese** (*Intermediate*)

REFEREES

Dr. Ammar Al-Bazi

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Dr. Viktor Dorfler

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